

plate only has 4, the remaining 8 are located on the motherboard and can be utilised by using header cables. The front panel audio and USB ports are actually connected to the headers. Basically, the most commonly used interfaces have a place on the backplate and the less common ones are accessed via headers. This is where you find legacy interfaces like floppy drive headers and COM port headers, both are a rare find. The headers are designed in a way that you can only insert the corresponding cable in one particular way, there will be a tab or a notch that makes it impossible to insert the cable anywhere else. One exception is the header that has contacts for the power LED, HDD activity LED, power and reset switches. Each manufacturer follows a different standard and cabinets have a bunch of cables instead of one unified cable for the same. Refer to your motherboard manual to figure out what goes where. For switches the polarity doesn't matter, i.e. you can insert the cable either way but for LEDs the + and - symbols must be followed.

All over the motherboard you will find small 3-pin or 4-pin headers, these are usually for connecting chassis cooling fans, the three pin headers don't support PWM while the 4-pin ones do. The 4-pin ones give your feedback from the fan by telling you what the current RPM is and you can increase the flow of air accordingly while the 3-pin fans are not localised and don't give greater control.

8. Expansion slots

Expansion slots are used to accommodate various cards that add extra features to your computer. Discrete graphic cards are the most commonly known example. The current standard is to use PCI (Peripheral Component Interconnect) and PCIE (PCI Express) slots. PCI slots are slowly being phased out as the processors can now afford greater bandwidth than the PCIE slots have and there are better circuits for managing bandwidth between multiple add-on cards. PCIE have slots based on the bandwidth needed, and thus we have x1 / x4 / x8 / x16 / x32 slots but the x32 slots are not to be found on motherboards. The following table lists the bandwidth supported by various slots.

Port	Bandwidth (Single direction)	Bandwidth (Dual direction)
x1	200 MBps	400 MBps
x4	800 MBps	1600 MBps
x8	1600 MBps	3200 MBps
x16	3200 MBps	6400 MBps